The genuinely eDMFT projector contribution to stress

Derivation, source audit, and independent tests

Working audit for Kristjan Haule

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Scope. This audit examines projector 5 and the current sources in `/Users/haule/projects/eDMFT_new/src/dmft2`. It does not repeat the Blaha G1/G2 derivations. Production sources and original material calculations were not changed. All new runs used isolated copies.

Main findings. The direct Matsubara gradient, matrix normalization derivative, and complementary-band response pass independent algebraic and finite-functional tests. One chain-rule inconsistency is established: the normalization overlap is symmetrized, but the adjoint symmetry operation in the normalization-window response is disabled by default. Applying the existing diagnostic switch changes the $SrVO_3$ mean stress by +0.1293 GPa and the FeSe mean stress by +0.1397 GPa; it does not resolve the hydrostatic discrepancy. Two FeSe shear checks fail and require a consistent equivalent-atom perturbation convention. A separate, more fundamental issue is potential dependence of a finite-band projector: ordinary charge self-consistency need not make that implemented functional stationary. An independent finite model demonstrates this effect; its size in the materials has not been measured.

1 Definitions and matrix orientations

The derivative ∂_ν refers to one homogeneous strain channel. The six channels are xx, yy, zz, yz, xz, xy. For a shear channel ab, a scalar perturbation t means $\epsilon_{ab} = \epsilon_{ba} = t/2$. Thus the derivative with respect to t is $\Omega\sigma_{ab}$, without an extra factor two. For isotropic length strain, $d\Gamma/dt = \Omega \operatorname{Tr}\sigma$ and the mean stress is $(3\Omega)^{-1} d\Gamma/dt$. Here the reported sign is $+\partial\Gamma/\Omega$.

Symbol	Definition
k, w_k	Reduced wave vector and normalized Brillouin-zone weight; $\partial_\nu w_k = 0$. Spin is an explicit index or a degeneracy factor in the functional sums, but not an extra factor in the overlap normalization.
χ_{Ik}	LAPW basis function; I, J label basis functions.
O_k, H_k	Basis overlap and KS Hamiltonian; $H_k A_k = O_k A_k \varepsilon_k$, $A_k^\dagger O_k A_k = 1$.
n, m, a	KS band indices. $n, m \in W_k$ when specified; $a \notin W_k$ denotes a complementary band.
α, β	Local correlated orbitals. Independent correlated blocks are summed; formulas below display one block for clarity.
ϕ_α	Chosen localized orbital, with its radial shape and local-axis convention specified.
$q_{I\alpha}$	$\langle\chi_I\|\phi_\alpha\rangle$, a basis-by-orbital matrix.
P_k	Raw band-by-orbital projector $A_{Wk}^\dagger q_k$.
S, X, U_k	Global overlap, $X = S^{-1/2}$, and normalized projector $U_k = P_k X$.

$\widetilde{\Sigma}$	$\Sigma_{\mathrm{imp}} - V_{\mathrm{DC}}$ in the correlated orbital basis.
$G_k(z)$	Band Green function, $[z - \varepsilon_k - U_k \widetilde{\Sigma}(z) U_k^\dagger]^{-1}$, with the self-energy embedded only in W_k; $z = i\omega_n + \mu$.
ρ_k, ρ_k^E	Density and energy-weighted density matrices with the same frequency summation, spin and k weights. ρ^E uses the regularized moment $zG - 1$. Both matrices are Hermitian after pairing frequencies.
C_k	Orbital-by-window-band functional gradient, defined in Eq. (5). It is not a projector and is not generally Hermitian.
$\mathcal{S}$	The linear symmetry reconstruction/averaging applied to the irreducible-k overlap. $\mathcal{S}^*$ is its adjoint under the real trace pairing.

All formulas assume a positive overlap on the active correlated subspace and a spectrally separated boundary of the fixed-index window. Neither condition requires U_k to have full column rank at every individual k.

2 Start from the functional, not the total energy

Write the stationary functional using independent variables $n, V, g, \widetilde{\Sigma}$:

$$\begin{aligned}\Gamma = {} & F_{\mathrm{sp}}[H_0 + V, \widetilde{\Sigma}, U] + E_{\mathrm{Hxc}}[n] - \int V n + E_{\mathrm{ion}} \\ & + \Phi_{\mathrm{imp}}[g] - \Phi_{\mathrm{DC}}[g] - \mathrm{Tr}(\widetilde{\Sigma} g).\end{aligned} \tag{1}$$

Here H_0 includes the kinetic operator and ionic external potential, V is the Hartree–XC potential, and F_{sp} is the regularized lattice $-T \sum \ln \det[-G^{-1}]$, including μN for a fixed-particle-number Helmholtz functional. The local trace includes frequency, block and spin sums. This is a functional before imposing stationarity, not a rearrangement of the nonstationary eDMFT total energy.

At the stationary impurity solution,

$$\frac{\delta(\Phi_{\mathrm{imp}} - \Phi_{\mathrm{DC}})}{\delta g} = \widetilde{\Sigma}, \qquad g = \sum_k w_k U_k^\dagger G_k U_k. \tag{2}$$

Consequently the $\widetilde{\Sigma}\,\delta g$ and $g\,\delta\widetilde{\Sigma}$ variations cancel between the local functional and the lattice term. A two-particle vertex is not needed for this first derivative. Explicit variation of U survives:

$$\delta\Sigma_{\mathrm{emb}} = \delta U \widetilde{\Sigma} U^\dagger + U \widetilde{\Sigma} \delta U^\dagger + U \delta\widetilde{\Sigma} U^\dagger, \tag{3}$$

$$D_\nu^{\mathrm{proj}} = 2\,\mathrm{Re} \sum_k \mathrm{Tr}[C_k\, \partial_\nu U_k], \tag{4}$$

$$C_k = w_k T \sum_{\omega_n} \widetilde{\Sigma}(i\omega_n) U_k^\dagger G_k(i\omega_n). \tag{5}$$

Spin sums or their degeneracy factors are included in C. The k weight must not be applied a second time. Negative Matsubara frequencies supply the conjugate partner needed for the real expression. A fixed self-energy is therefore sufficient to test this term, with the impurity functional replaced by its tangent plane. This leaves F_{sp} plus the density/ionic terms and a strain-independent constant; it does not eliminate the projector chain.

In a moving nonorthogonal basis the full one-particle variation is

$$D_\nu^{\mathrm{1p}} = \sum_k \mathrm{Re}\,\mathrm{Tr}[\rho_k H_{k,\nu} - \rho_k^E O_{k,\nu}] + D_\nu^{\mathrm{proj,explicit}}. \tag{6}$$

The meaning of "explicit" must match the representation: once KS-state connections have been absorbed into the matrix pair, those same connections cannot be added a second time through $\partial_\nu U$.

3 The direct gradient and the $B = UC$ identity

Dyson's equation gives, with a consistent regularization,

$$B_k \equiv \rho_k^E - \varepsilon_k \rho_k = U_k C_k. \tag{7}$$

The KS energy matrix multiplies ρ on the *left*. B need not be Hermitian. Recovering C by a pseudoinverse of U can lose its component in $\ker U$. The current production code instead accumulates C directly. For a positive Matsubara frequency its paired integrand is

$$C^{\text{pair}}(i\omega) = \widetilde{\Sigma} U^\dagger G + \widetilde{\Sigma}^\dagger U^\dagger G^\dagger. \tag{8}$$

The second term is not the adjoint of the first rectangular matrix. `AccumulateDirectProjectorPair` implements these products in this order. Its static-reference and high-frequency corrections are assembled with the same conventions as ρ^E. Tests differentiate a complex log determinant and include a rank-deficient projector; both pass. Actual $SrVO_3$ and FeSe diagnostics also compare B against an independently accumulated $\Sigma_{\text{emb}} G$ moment.

4 Differentiate the actual matrix normalization

The implemented overlap and normalized projector are

$$S = \mathcal{S}\left[\sum_k w_k P_k^\dagger P_k\right], \qquad X = S^{-1/2}, \qquad U_k = P_k X. \tag{9}$$

Thus

$$\partial_\nu U_k = (\partial_\nu P_k) X + P_k (\partial_\nu X), \tag{10}$$

$$\partial_\nu S = \mathcal{S}\left[\sum_k w_k ((\partial_\nu P_k)^\dagger P_k + P_k^\dagger \partial_\nu P_k)\right]. \tag{11}$$

For a symmetry-preserving strain the symmetry map is fixed. For general strain the Cartesian derivative indices must be transformed as well; one must use the adjoint of that complete map, not assume independent scalar symmetrization of each shear component.

Put $Y = X^{-1}$. Differentiating $XSX = 1$ gives the Sylvester equation

$$Y(\partial_\nu X) + (\partial_\nu X) Y = -X(\partial_\nu S) X. \tag{12}$$

If $X = V_x \operatorname{diag}(x_i) V_x^\dagger$, primes on matrices in the next equation mean transformation to this eigenbasis:

$$(\partial_\nu X)'_{ij} = -\frac{x_i x_j (\partial_\nu S)'_{ij}}{1/x_i + 1/x_j}, \qquad (X^{-1}\partial_\nu X)'_{ij} = -\frac{x_j (\partial_\nu S)'_{ij}}{1/x_i + 1/x_j}. \tag{13}$$

No derivative of the eigenvectors of X is required. In particular, $\partial_\nu X = -\frac{1}{2} S^{-3/2} \partial_\nu S$ is invalid unless the matrices commute.

An economical way to carry the normalization response is its adjoint. Define the Hermitian orbital matrix and the linear Sylvester map

$$J = \mathrm{Herm} \sum_k C_k P_k, \qquad \mathcal{L}_X[\delta S] = \delta X, \qquad Q = \mathcal{L}_X[J]. \tag{14}$$

$\mathcal{L}_X$ is self-adjoint on Hermitian matrices. Therefore

$$D_\nu^X = 2\,\mathrm{Tr}[J \partial_\nu X] = 2\,\mathrm{Tr}[Q \partial_\nu S] \tag{15}$$

$$= 2\,\mathrm{Tr}\left[\underbrace{\mathcal{S}^*[Q]}_{Q_{\mathrm{eff}}} \sum_k w_k \left((\partial_\nu P_k)^\dagger P_k + P_k^\dagger \partial_\nu P_k \right) \right]. \tag{16}$$

An especially useful complete gradient with respect to the raw projector is

$$\boxed{\mathcal{C}_{P,k} = X C_k + 2 w_k Q_{\mathrm{eff}} P_k^\dagger, \qquad D_\nu^{\mathrm{proj}} = 2\,\mathrm{Re} \sum_k \mathrm{Tr}[\mathcal{C}_{P,k} \partial_\nu P_k].} \tag{17}$$

The source stores the equivalent moment $(\sum_k C_k U_k)^T$ and uses $X^{-1}\partial_\nu X$. Its conjugations agree with the above formula; comments using a band-by-orbital Frobenius gradient use a different orientation of C.

Confirmed chain-rule omission. The forward overlap is symmetrized, while `Symmetrize NormalizationMoment` is default-off. In the symmetry-invariant case, projecting the moment before the Sylvester adjoint is equivalent to Eq. (16). In general the robust construction is to apply the exact adjoint after it, respecting active blocks and the order of equivalent-atom averaging. The diagnostic switch `EDMFT_STRESS_NORM_SYMMETRY=APPLY` tests the existing implementation; changing that default alone is not a proof of all general shear conventions.

5 Fixed band indices still have a window derivative

Let A_n be a normalized generalized KS eigenvector. In the reference band basis define $T_{an} = A_a^\dagger O \partial_\nu A_n$. For separated energies,

$$T_{an} = \frac{H_{\nu,an} - \varepsilon_n O_{\nu,an}}{\varepsilon_n - \varepsilon_a} \quad (a \neq n), \qquad T_{nn} = -\tfrac{1}{2} O_{\nu,nn}, \tag{18}$$

where the diagonal imaginary gauge is chosen to vanish. Then

$$\partial_\nu P_W = T_{F,W}^\dagger P_F + A_W^\dagger \partial_\nu q. \tag{19}$$

F is the full available band set, not just the window. There is no moving energy-cutoff delta function for fixed band indices. Nevertheless, the selected spectral subspace moves, and its complementary bands enter Eq. (18). At a degeneracy crossing a window edge, the spectral projector is not differentiable without a prescription that includes the complete degenerate subspace. Silently skipping a small denominator is not a physical resolution of that problem.

At fixed X, put $K_{an} = (U_F C)_{an}$. The outside-window contribution is

$$D_\nu^W = 2\,\mathrm{Re} \sum_k \sum_{n \in W_k, a \notin W_k} \frac{K_{an} \left[H_{\nu,an} - \varepsilon_n O_{\nu,an} \right]^*}{\varepsilon_n - \varepsilon_a}. \tag{20}$$

It is represented by Hermitian response matrices

$$R_{an} = K_{an}/(\varepsilon_n - \varepsilon_a), \qquad R_{na} = R^*_{an}, \tag{21}$$

$$R^E_{an} = \varepsilon_n R_{an}, \qquad R^E_{na} = (R^E_{an})^*, \tag{22}$$

$$D^W_\nu = \mathrm{Re}\sum_k \mathrm{Tr}[R_k H_{k,\nu} - R^E_k O_{k,\nu}]. \tag{23}$$

Putting K_{an} in the transposed matrix slot breaks complex KS-phase invariance. The current source has the correct orientation and passes a rephasing test.

The normalization window response uses the same construction with the raw projector and $C^N_{P,k} = 2w_k Q_{\mathrm{eff}} P^\dagger_k$. It also needs the within-window metric term

$$\partial_\nu S_{\mathrm{metric}} = -\mathcal{S}\sum_k w_k P^\dagger_W O_{\nu,WW} P_W. \tag{24}$$

The source includes it through $R^{E,N}_{WW} = \mathrm{Herm}(P_W C^N_{P,k})$. Internal unitary rotations cancel from $P^\dagger_W P_W$; the metric term does not. The remaining within-window connection is already contained in Eq. (6), using $B = UC$. Adding it again as a separate correction would double count it.

6 Raw projector 5, radial response, and the sphere boundary

Schematically the local raw projector is

$$P_{n\alpha} = \sum_\lambda a^*_{\lambda n} I_{\lambda\alpha} + P^{\mathrm{interstitial}}_{n\alpha}, \qquad I_{\lambda\alpha} = \int_0^{R_{\mathrm{MT}}} r^2 u_\lambda(r)^* \phi_\alpha(r)\, dr. \tag{25}$$

Angular, spin and local-axis transformations are understood. The primitive index λ includes $u_l, \dot{u}_l$ and all local orbitals. Consequently

$$\partial_\nu P_{n\alpha} = \sum_\lambda (\partial_\nu a_{\lambda n})^* I_{\lambda\alpha} + \sum_\lambda a^*_{\lambda n} \partial_\nu I_{\lambda\alpha} + \partial_\nu P^{\mathrm{interstitial}}_{n\alpha}. \tag{26}$$

The production builder includes matching/Bessel changes, angular derivatives, the $-P/2$ volume term in each diagonal channel, and the projector-5 interstitial continuation when present. It also has a branch for $\partial_\nu I$, but `drix_mat` is zero in the normal production path. That is a frozen-radial approximation, not proof that $\partial_\nu I = 0$ along a charge-relaxed strain path. Its repair must be paired with the same radial response in H_ν and O_ν.

A projector finite at R_{MT} does not automatically imply a new independent surface correction. With fixed physical radius and coordinates relative to the co-moving atom, the integration domain is fixed. If the radius changes, there is explicitly

$$(\partial_\nu I)_{\mathrm{boundary}} = R^2_{\mathrm{MT}} (\partial_\nu R_{\mathrm{MT}}) u_\lambda(R_{\mathrm{MT}})^* \phi_\alpha(R_{\mathrm{MT}}), \tag{27}$$

with the angular contraction included as appropriate. A laboratory-coordinate derivation must combine moving-boundary and translation terms consistently. For a projector continued into the interstitial, the inside/outside boundary pieces must be combined before claiming an additional surface term.

7 A stationarity issue specific to spectral projectors

This section is a separate derivation, not a claim that the observed GPa error has already been explained. In a fixed Hilbert space define

$$\Pi_W[V] = \sum_{n\in W} |\psi_n[V]\rangle\langle\psi_n[V]|, \qquad |\Phi_W[V]\rangle = \Pi_W[V]|\phi\rangle X[V]. \tag{28}$$

The finite-window embedding is a physical function of V, even when the radial orbital on disk is fixed. A basis change alone is different and cancels exactly between the terms in Eq. (6).

Varying Eq. (1) at fixed geometry gives

$$\delta_V\Gamma = \int (n_G - n)\delta V + 2\,\mathrm{Re}\sum_k \mathrm{Tr}[C_k\delta_V U_k]_{\text{physical window}}. \tag{29}$$

Ordinary charge self-consistency sets $n = n_G$. It does not set the second term to zero. At fixed basis, $\delta O = 0$, $\delta q = 0$, this residual is

$$\delta_V\Gamma = \mathrm{Re}\sum_k \mathrm{Tr}[(R_k + R_k^N)\delta V_k]. \tag{30}$$

The matrices are exactly the complementary-window and normalization-window responses above. The inspected code uses them in force/stress contractions, but constructs the charge from the ordinary lattice density matrix.

It follows that a derivative along a converged charge path may contain

$$\boxed{D_\nu^{\text{path}} = D_\nu^{\text{explicit}} + \mathrm{Re}\sum_k \mathrm{Tr}[(R_k + R_k^N)(\partial_\nu V_{\text{Hxc}})_k] + \text{other projector/basis response terms.}} \tag{31}$$

Here the extra V derivative means the induced potential variation excluded from the explicit strain convention, not a second copy of a potential derivative already present in H_ν. The exact split must be verified against the LAPW stress kernels. Complete stationary variation would instead require the projector contribution in the V stationarity equation, or a projector definition genuinely independent of V. Either choice changes the formulation and must not be made silently as a stress-only patch.

An independent dynamic-self-energy finite model verifies Eq. (29) without radial approximations or core/valence cancellation:

Test	Finite W	Complete space
$\delta F_{\text{sp}} - \mathrm{Tr}(\rho\delta V)$	0.035133243630	$-6.5\,10^{-11}$
Predicted window chain	0.035133243587	0

These are model energy-derivative units, not GPa. The density equals the exact lattice density at the reference point. Thus the residual is not an SCF tolerance error. The material value of Eq. (31) remains unmeasured, and a full-space model does not establish completeness of a finite LAPW basis.

8 Tests and implementation status

The regression harness extracts production Fortran routines and compiles them with bounds checking. Independent Python endpoints diagonalize the generalized Hamiltonian and normalization matrices. The joint test uses three unequal k weights, complex projectors, both lower and upper complementary bands, nonzero H_ν, O_ν, q_ν, and noncommuting matrices. A second version uses an exactly soluble dynamical self-energy $\widetilde{\Sigma}(z) = \Sigma_\infty + V_b(z - E_b)^{-1}V_b^\dagger$; its free energy is obtained from the enlarged bath Hamiltonian without a Matsubara-tail approximation.

Test	Result
Inverse square root against endpoint diagonalization	Pass, $4.7\,10^{-12}$
Normalization adjoint	Pass, $1.6\,10^{-11}$
Complex window response and KS rephasing	Pass
Generalized H_ν/O_ν window contraction	Pass, $1.5\,10^{-11}$
Within-window normalization metric	Pass, $3.4\,10^{-11}$
Pure basis-connection cancellation	Pass, $3.5\,10^{-17}$
Direct C, paired Matsubara / real axis	Pass, below $3\,10^{-11}$
Rank-deficient U with direct C	Pass, $2.1\,10^{-11}$
All projector chains, no symmetry map	Pass, below 10^{-10}
All chains, symmetrized S, default response	Fail
Same test with adjoint symmetry	Pass, below 10^{-10}
Potential stationarity with a finite window	Nonzero residual reproduced
Same stationarity test with the complete space	Residual below 10^{-10}

For example, in the joint dynamical test with symmetrized S, the finite derivative is -0.039320865142. The default chain gives -0.099349880276; the adjoint-complete chain gives -0.039320865133. This proves a chain-rule failure for the tested map, without fitting a stress correction to a material. The failure persists with a symmetry-preserving dynamical self-energy: finite difference -0.105243690363, default chain -0.105612238523, adjoint-complete chain -0.105243690327. It is therefore not an artifact of testing a symmetry-breaking self-energy against a symmetrized overlap.

Actual material checks

Check	$SrVO_3$	FeSe RELA
Maximum diagonal raw-projector relative error	$5.81\,10^{-10}$	$1.15\,10^{-9}$
Maximum shear raw-projector relative error	$7.59\,10^{-10}$	0.5802
Symmetry-adjoint mean-stress change (GPa)	+0.1293	+0.1397

The switch changes the normalization-window response. Neither the impurity self-energy nor the charge inputs were changed. SUM, VSUM, YSUM and DRHO are identical between the two modes in each material. The $SrVO_3$ hydrostatic mismatch remains; the switch slightly increases it. These are changes in *mean* stress, already divided by three, with the final mixer sign.

The independently accumulated $\Sigma_{\mathrm{emb}}G$ moment agrees with $\rho^E - \varepsilon\rho$ to maximum relative errors $6.82\,10^{-14}$ in $SrVO_3$ and $3.46\,10^{-14}$ in FeSe. Direct C and its pseudoinverse reconstruction agree to $2.22\,10^{-14}$ and $1.19\,10^{-14}$, respectively, in these runs. Thus a lost pseudoinverse component is not the current material discrepancy. A separate rank-deficient model still requires direct C, and the production code correctly retains it.

Reusing the saved central density and core contributions, the fresh valence runs give mean stresses $5.54758 \to 5.67688$ GPa in $SrVO_3$ and $-0.25286 \to -0.11317$ GPa in FeSe when the switch is applied.

The previously audited numerical references are about 2.934 and -1.0491 GPa, respectively. The symmetry step therefore slightly worsens both comparisons. These small source/assembly checks are not newly reconverged strain triplets.

All four full material runs completed and the audited outputs were finite. Their Fortran termination messages included sticky invalid/divide-by-zero IEEE flags; the origin of those flags was not traced in this pass.

The raw-projector checks hold eigenvector coefficients and radial functions fixed, use the first irreducible k point, and perturb geometry by 10^{-6}. They test the explicit builder, not the complete relaxed projector. For FeSe, the failed xz and xy channels are exactly those reversed by the operation $\mathrm{diag}(1,-1,-1)$ relating the two Fe atoms. The builder's matching magnitudes use shared Bessel arrays computed from $|B(k+G)|$, whereas some analytic factors use $|BR(k+G)|$. These agree at the symmetric reference and under diagonal strain, but differ under the symmetry-breaking shears. The finite-difference helper retains the original equivalence map. This identifies a convention inconsistency to isolate; it does not by itself decide whether the physical shear formula or the restricted helper should be changed. It is not an explanation of the hydrostatic discrepancy.

Separate scratch calculations retaining only the first or only the second correlated Fe atom confirm the localization. Atom 1 passes every channel with maximum relative error $1.39\,10^{-9}$. Atom 2 has relative errors 0.7148 and 0.7805 in xz and xy, respectively; its other four channels pass at about 10^{-9}. The absolute discrepancy norms are exactly the same as in the two-atom test. These one-k probes test projector geometry only and are not meaningful material pressure calculations.

9 Source map and the next controlled tests

Routine / file	Audited role
`cmp_dmft_weights.f90`	Direct C, positive/negative-frequency pairing, energy moments, static reference and tails.
`Build_DMFT_Projector` in `l2mainSubs.f90`	Explicit raw projector, matching/angular/volume terms; optional ∂I.
`cmp_overlap`	k-summed raw overlap and explicit derivative, symmetry reconstruction, X, preservation of a saved normalization.
`ComputeNormalizationDerivative`	Sylvester derivative, Eq. (12).
`BuildWindowResponseMatrices`	Direct C selection and fixed-X complementary-band matrices.
`AccumulateNormalizationMoment`, `BuildNormalizationAdjoint`	Global k-coupled normalization gradient.
`BuildNormalizationWindowResponseMatrices`	Complementary normalization response plus the within-window metric.
`symmetrize.f90`	Forward overlap symmetry; adjoint-moment switch defaults to off; Cartesian derivative rotations and equivalent-block averaging.
`l2main.f90`	Calls the above, contracts $2\,\mathrm{Re}\,C\partial U$, and applies the negative sign of the intermediate WIEN `STR_VAL` convention.
`p_project.f90`	Projector radial overlaps; production `drix_mat=0`.

Recommended next work, requiring approval before source edits:

1. Implement the exact adjoint of the actual normalization/symmetry map, including the order of equivalent-block averaging. Keep a regression test with noncommuting complex matrices and a dynamical self-energy. Its necessity is mathematical; the sign must not be chosen to improve a material number.

2. Isolate the FeSe shear check by equivalent atom and separate Bessel, angular and LO pieces. For broken symmetry use a consistent full-cell perturbation or rotate the strain together with the atom mapping.

3. At one fixed $SrVO_3$ geometry, perturb the input density/potential in a charge-neutral direction and evaluate the frozen-self-energy tangent functional. Compare its slope with $\mathrm{Tr}[(R + R^N)\delta V]$ using the same potential matrices. Include a DFT control to separate radial-basis effects. This directly tests the stationarity gap; it avoids subtracting large independent strained core and valence energies.

4. Only then contract the measured induced V_{Hxc} response along the strain path. Establish which part is already in the existing kernels before adding Eq. (31) to any reported stress.

5. Complete the paired radial $H/O/P$ response and repeat stored-band convergence. Stored complementary bands remain an approximation; increasing their number does not enlarge the chosen DMFT window itself.

Other limitations remain: window-edge degeneracies, local-axis conventions with a diagonal-only self-energy, an outdated saved normalization, and consistency of active-orbital selection. The forward normalization uses nonzero diagonal `Sigind` entries, while some response routines accept any active row/column; a purely off-diagonal active block is therefore an additional unsupported corner case. It is not the present $SrVO_3$/FeSe setup.

An adjacent force-path limitation also deserves a separate audit: the global normalization-window pass is gated by `Qstress`, whereas the fixed-X window response is also used for forces. Explicit force normalization derivatives exist, but this does not establish that the full normalization-window force chain is present. No new force benchmark was performed in this pass.

Reproducibility. The accompanying Python scripts and JSON results record the combined functional test, potential-stationarity test, material diagnostics, source hashes, and copies of the small diagnostic outputs. They do not change the production executable or commit any source changes.